

UNIT-V

19. Illustrate the Lossy Compression Algorithms.

Or

20. How to convert a colour image to a grey scale image and vice versa? Describe the different Colour Models in detail.

5634084

B.Tech. DEGREE EXAMINATION, JULY 2023.

Fourth Semester

Computer Science Engineering

GRAPHICS AND IMAGE PROCESSING

(2013-14 Regulations)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. List the operating characteristics of Plasma Panels.
2. What is an aspect ratio in display?
3. Define clipping.
4. Write a general transformation matrix for shearing in x-direction.
5. State the goals of Digital Image Processing.
6. What is the difference between Image Convolution and Correlation? Is there any difference between them in image Processing?
7. Is there any advantage of frequency domain filters over spatial filters? Why?

8. Differentiate between Image Enhancement and Image Restoration.

9. What is meant by Tonal and Colour Correction?

10. Write the role of Morphology in image Processing.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions

UNIT-I

11. Narrate the different types of Video Display Devices.

Or

12. Discuss on the Interactive Picture Construction Techniques.

UNIT-II

13. Explain in detail the Cohen-Sutherland line clipping algorithm with an example.

Or

14. Derive the decision parameter expressions for Bresenham line drawing algorithm. Write Bresenham line drawing algorithm and explain how it is better than DDA algorithm for line generation.

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UNIT-III

15. (a) Write a brief survey on Image Processing Applications. (5)

(b) Expound the Digital Imaging Systems and its types. (6)

Or

16. Enumerate the various Image Processing Operations also explain the Arithmetic and Logical Operations with examples.

UNIT-IV

17. Analyze the importance of Histogram-based Image Enhancement Techniques and Perform histogram equalization on the following Images:

$$\begin{bmatrix} 1 & 3 & 5 \\ 4 & 4 & 3 \\ 5 & 2 & 2 \end{bmatrix}$$

Or

18. Examine the Constrained and Unconstrained methods of Image Restoration.

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